The Productivity Imperative in the Face of the Demographic Winter: An Analysis of the Spanish Economy under the Prism of the Solow Model (2025-2050)

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Abstract

The economic trajectory of Spain is at a historical juncture defined by an unprecedented demographic transformation. After decades of growth based on integration into international markets and factor accumulation, the country now faces a phenomenon that threatens to redefine the laws of its potential growth: the accelerated aging of its population and a birth rate that is systematically among the lowest in the developed world. This report addresses this issue through the rigorous application and empirical calibration of the Solow growth model, seeking to determine whether technological progress and capital deepening can compensate for the projected contraction of the labor force.

1 Formulation of the Economic Question and Justification of its Relevance

The central question guiding this research is as follows: **To what extent can total factor productivity growth and physical capital intensification mitigate the negative impact of demographic decline on Gross Domestic Product (GDP) per capita growth in Spain during the 2025-2050 period?**

Answering this question is not merely an academic exercise, but an existential necessity for the sustainability of the Spanish social and economic model. The relevance of this analysis is based on three critical pillars that define the vulnerability of the national economy to demographic change.

First, the speed of aging in Spain is higher than most of its OECD peers [1]. While other advanced economies have had decades to adapt, Spain faces a concentrated demographic "shock" in which the baby boom generation is simultaneously entering retirement age.

Second, the sustainability of public finances is directly linked to the support or dependency ratio. According to data from the National Statistics Institute (INE), the Spanish population over 65 years old, which currently represents 20.4% of the total, will reach a maximum of 30.5% around 2055 [4]. This inversion of the population pyramid exerts immense fiscal pressure on the pension and health systems, whose financing falls on an increasingly reduced cohort of workers.

Finally, real income convergence with Europe has stalled. Despite resilient post-pandemic growth, GDP per hour worked in Spain remained 7% lower than the average of the European Union in 2024, suggesting that the Spanish economy does not fully leverage technological progress to compensate for labor shortages [6].

The importance of this study lies in the fact that, through the Solow model, we can isolate the variables of savings, depreciation, and technological growth to understand whether current policies are sufficient to avoid secular stagnation. The analysis of these dynamics allows us to identify whether Spain is heading towards a scenario of "capital intensification," where each worker is more productive because they have better tools and technology, or if the weight of dependency will end up draining national savings and, therefore, productive investment.

2 Macroeconomic Model Specification: The Solow Framework

To address this issue, the Solow growth model (1956) has been selected, a fundamental pillar of neoclassical theory that allows decomposing economic growth into the contributions of capital, labor, and technological progress.

Structure and Assumptions of the Model

The Solow model describes a closed economy where a single homogeneous good is produced using capital (K) and labor (L) under a technology that progresses exogenously (A). The aggregate production function is usually specified using a Cobb-Douglas form, which presents constant returns to scale and diminishing returns for each factor individually:

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha} \tag{1}$$

Where Y_t is the total output at time t, α is the elasticity of output with respect to capital (or the capital share in national income), and $1-\alpha$ is the labor share. The term A_t represents labor-augmenting technological progress (Harrod neutrality), which ensures that the model can reach a balanced growth path.

The model assumes that a constant fraction s of income is saved and invested ($I_t = sY_t$), and that capital depreciates at a constant rate δ. The law of motion for capital is:

$$K_{t+1} = (1-\delta)K_t + sY_t \tag{2}$$

To adapt the model to the demographic issue, we define the growth rates of the population (n) and technology (g) as:

$$L_{t+1} = (1+n)L_t$$
$$A_{t+1} = (1+g)A_t$$

The Intensive Transformation and the Steady State

To analyze per capita welfare, we transform the model to its intensive form, defining capital per unit of effective labor as $k = K/(AL)$. The dynamics of capital in terms of efficiency are given by the fundamental Solow equation:

$$k_{t+1} = \frac{sf(k_t) + (1-\delta)k_t}{(1+n)(1+g)} \tag{3}$$

In the steady state ($k_{t+1} = k_t = k^*$), investment per unit of effective labor must be exactly equal to the "break-even" investment necessary to compensate for depreciation, population growth, and technological progress:

$$s(k^*)^\alpha = (n + g + \delta + ng)k^* \tag{4}$$

This relationship is crucial for answering our economic question. If the population growth rate (n) decreases or becomes negative (as projected for the native segment in Spain), the break-even investment requirement falls, theoretically allowing capital per worker (k^*) to increase. This phenomenon is known as **capital deepening**. However, adapting the model for Spain must consider whether this theoretical gain is real or if it is nullified by other structural factors.

3 Adaptation of the Model to the Spanish Context

Adapting the Solow model for Spain does not only consist of changing parameter values, but of interpreting what each variable means in the current context of reforms and challenges.

First, the variable n (population growth) must be interpreted as the growth of the *effective labor force*. In Spain, there is a massive divergence between total population growth and working-age population growth. While migration is sustaining the total population, the productive native cohort is shrinking rapidly [5]. Therefore, we adapt the model using a rate n that reflects the expected moderation of migratory flows and the impact of aging on the activity rate [2].

Second, the savings rate (s) cannot be considered totally independent of demographics. According to the life-cycle hypothesis, an aging population tends to dissave to finance consumption in retirement [9]. Furthermore, the increase in public spending on pensions and health (which AIReF projects will rise from 12.7% to 16.1% of GDP by 2050 in the case of pensions) may reduce national savings if not compensated by budget surpluses or structural reforms [3]. Therefore, the model is analyzed under different assumptions of s to capture the possible "crowding out" effect of aging spending on productive investment.

Finally, technological progress (g) is adapted considering the Spanish "productivity challenge". Unlike economies at the technological frontier (such as the US or Germany), TFP growth in Spain has historically been low and volatile [6]. Adapting the model involves calibrating g not as a generic value, but as the productivity growth target necessary to maintain per capita income in the face of falling L.

4 Calibration of the Model for the Spanish Economy

Calibration is the process of assigning values to the model parameters so that it is consistent with observed data and long-term projections of the Spanish economy.

Capital Share in Income (α): Determining α is critical, as it defines the returns to capital accumulation. In standard macroeconomic literature, a value of 0.33 is often used. However, the Spanish productive structure has particularities. Specific studies by the Bank of Spain and analyses based on the Penn World Tables suggest that the labor share in Spain is around 62.6%, implying an α of 0.374 [10]. Other works breaking down capital between equipment and structures find combined shares reflecting the intensive use of infrastructure in the Spanish growth model [7]. For this analysis, a value of **0.37** is adopted, consistent with the factorial distribution of income reported by national institutions for the post-pandemic period.

Depreciation Rate (δ): Depreciation reflects the wear and obsolescence of physical capital. Calibrating this parameter must consider the asset mix in the Spanish economy. While software and machinery depreciate rapidly (rates above 15%), non-residential structures and housing do so much more slowly (between 1.3% and 2%) [7]. Weighted average calculations for the Spanish economy, based on INE and Eurostat data, place the aggregate annual depreciation rate at **3.8%** ($\delta = 0.038$) [10]. This value is slightly lower than the OECD average, reflecting the significant weight of capital in structures within the total Spanish capital stock.

Savings and Investment Rate (s): In the Solow model, savings translates directly into productive investment. To calibrate s, we observe gross fixed capital formation as a percentage of GDP. The World Bank and INE report that investment in Spain stood at 20.47% in 2024, having fallen from levels near 30% before the 2008 financial crisis. The total gross savings rate of the economy has recovered to 24.6% in mid-2025 [12]. For a conservative long-term calibration, we establish $s = \mathbf{0.21}$, a value reflecting the structural investment capacity of the economy once cyclical effects stabilize [11].

Productivity Growth (g): Exogenous technological progress (g) in the Solow model is the only driver of long-term per capita income growth. In Spain, total factor productivity (TFP) growth has been a historical weakness. However, AIReF and IMF projections for the 2025-2050 horizon assume that structural reforms and digitization could boost labor productivity at an average rate of 1.1% annually [3]. We calibrate $g = \mathbf{0.011}$, representing an optimistic but necessary scenario for convergence.

Population Growth (n): This is the central parameter of our study. According to the INE, although the total population will grow due to migration, the native labor force will contract. The US Census Bureau projects that Spain's population growth rate will tend towards negative values, reaching a minimum of -0.42% by 2050 [12]. However, considering the expected migratory flow that the Government and the Bank of Spain consider essential, we establish a steady-state rate n of **-0.001** (-0.1%), reflecting a labor force that, at best, remains stagnant or in slight decline [8].

5 Analysis Argumentation and Research Perspectives

Once the model is calibrated, we can perform a deep qualitative analysis of the implications of these values for Spain's future. The calibration reveals that Spain possesses a relatively high capital share (α), suggesting the economy is very sensitive to physical capital accumulation [7].

The Mechanism of Capital Intensification

Under the Solow framework, a reduction in n (as Spain is experiencing) reduces the slope of the required investment line ($n + g + \delta$). This means that, for a given level of savings s, the economy can sustain a higher level of capital per worker in the steady state. Theoretically, demographic aging "forces" the economy to be more capital-intensive: facing a labor shortage, investment shifts to machines and technology. This aligns with the view that AI and automation are the natural response to labor scarcity.

However, this report identifies a critical risk: if population aging reduces the national savings rate (s) due to pension and dependency spending, the positive effect of the fall in n could be nullified [3]. If fiscal pressures force a reduction in public investment to cover social transfers, Spain could fall into a low-growth trap.

Productivity as the Only Sustainable Engine

The calibration of g at 1.1% is the most uncertain and decisive factor. The Solow model demonstrates that in the steady state, per capita output growth is identical to g. This implies that if Spain fails to raise its productivity to the projected rate, citizen welfare will inevitably stagnate, regardless of capital accumulation. The model adaptation suggests Spain's success depends not on having more workers, but on existing workers operating at the technological frontier [6].

6 Conclusions of the Calibration Framework

The analysis allows for powerful conclusions for Spanish economic policy. First, Spanish demographics are in a phase where the decline of the native active population is a parametric reality [2]. A negative or near-zero n rate is the new natural state. Second, the calibration of α at 0.37 indicates that physical investment still has a significant return, but vulnerability lies in the savings rate; if aging drains resources for investment, the economy cannot reach the necessary capital per worker level [3]. Finally, the answer to the original question is conditional: demographic decline can be compensated, but only if Spain stabilizes its savings rate above 20% of GDP and simultaneously raises TFP growth to 1.1% annually. Without this technological boost (g), the increase in capital per capita resulting from the fall in n will have diminishing returns.

7 Simulation and Dynamic Analysis

Having calibrated the model to the specifics of the Spanish economy ($s = 0.21, \alpha = 0.37, g = 1.1\%$), we proceed to simulate the impact of the demographic winter. The

objective is to quantify the theoretical response of the main macroeconomic aggregates to a permanent structural change in the population growth rate (n).

Implementation and Shock Specification

The model was solved using Dynare under a deterministic simulation framework (perfect foresight). We assume the economy starts from a steady state with a stable population ($n = 0$) and is hit by a permanent demographic shock where the growth rate of the working-age population turns negative. Specifically, based on the projections discussed in Section 3, we simulate a transition where n falls from 0% to -0.1% ($n = -0.001$). The simulation horizon covers 100 periods (years).

Predictions and Dynamics

Figure 1 presents the dynamic adjustment of the economy following the demographic shock. The results confirm the neoclassical mechanism of *capital deepening*.

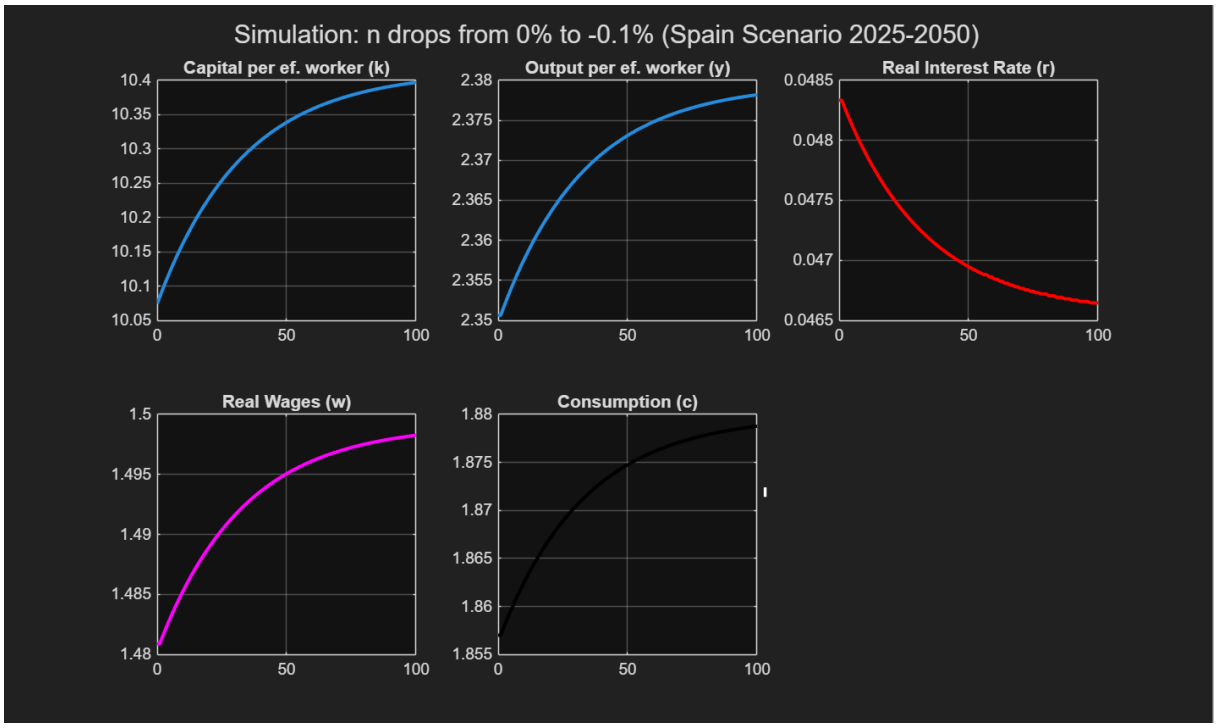

Figure 1: Dynamic response to a demographic shock (n falls from 0% to -0.1%).

The simulation yields the following predictions:

- **Capital Intensification (k):** Capital per effective worker rises significantly, transitioning from approximately 10.05 to a new steady state of 10.35. With fewer new workers, the "break-even investment" decreases.
- **Decline in the Real Interest Rate (r):** There is a secular decline in the real interest rate, falling from 4.85% to 4.67%. As capital becomes more abundant relative to labor, its marginal productivity diminishes ($\alpha k^{\alpha-1}$).
- **Increase in Wages and Output (w, y):** Labor scarcity and higher capital intensity increase the marginal productivity of workers, raising real wages (from 1.48 to

1.50) and output per effective worker.

Interpretation and Answer to the Research Question

The simulation answers the central question: **Yes, capital intensification can mitigate demographic decline, but with implications for returns.** The "demographic winter" acts as a driver of labor productivity growth through the capital deepening channel. However, this comes at a cost for savers, validating the hypothesis of secular stagnation in returns. For Spain, this implies that wealth accumulation will generate lower passive returns, structurally altering income distribution in favor of labor income over capital income.

8 Model Limitations and Future Research Lines

Although the Solow model provides a solid theoretical framework for understanding the mechanics of capital intensification, it is necessary to acknowledge certain structural simplifications that limit the predictive scope of this analysis, particularly in the context of a complex economy like Spain's.

The Rigidity of the Savings Rate (s)

One of the most significant limitations of this simulation is the assumption of a constant savings rate ($s = 0.21$). The Solow model treats savings as an exogenous fraction of income. However, the Life Cycle Hypothesis suggests that demographic aging has a direct impact on the propensity to save: retirees tend to "dissave" (consume assets) to finance their consumption.

In the Spanish reality, an increase in the dependency ratio could exert downward pressure on the aggregate savings rate, an effect that our model does not endogenously capture. If s were to fall simultaneously with n, the positive effect of capital intensification (k) would be smaller than estimated.

The Exogeneity of Technological Progress (g)

The model demonstrates that long-run growth in living standards depends almost exclusively on g (calibrated at 1.1%). However, in the Solow framework, technology is exogenous ("falls from the sky"). The model does not explain *how* this growth is achieved, nor does it consider whether population aging might negatively affect innovation (for instance, if an older workforce were less dynamic in adopting new technologies). If the demographic winter were to reduce g, the optimistic predictions regarding per capita GDP would be compromised.

Closed Economy Assumption

The standard calibration of the Solow model assumes a closed economy where Investment equals National Savings ($I = S$). Spain, as a member of the European Monetary Union, is an open economy with free capital mobility. This implies that a scarcity of domestic savings could be compensated by foreign capital flows, or that the decline in the return

on capital (r) could trigger capital flight toward younger and more profitable economies, potentially neutralizing the capital intensification process simulated here.

Absence of Human Capital and Heterogeneity

Finally, the model treats the labor factor (L) as a homogeneous input. It does not distinguish between young workers (in the learning phase) and senior workers (with greater experience but perhaps lower physical productivity), nor does it capture improvements in human capital (education). In a scenario where the labor force is contracting, the quality (skills) of the remaining workers becomes as critical as the quantity, a variable that the simple aggregate Cobb-Douglas production function does not allow us to disaggregate.

9 Concluding Remarks

This study has addressed the critical question of whether the Spanish economy can sustain its standard of living in the face of an unprecedented demographic contraction between 2025 and 2050. Through the calibration and simulation of the Solow model, we have identified that the "demographic winter" triggers an automatic mechanism of *capital deepening*. As the labor force shrinks, the remaining capital per worker increases, theoretically boosting labor productivity (y) and real wages (w).

However, this theoretical optimism must be tempered by the structural limitations identified. While the model predicts that lower population growth (n) can be compensated by higher capital intensity (k), this result is strictly conditional on the stability of the savings rate (s) and the realization of technological progress (g).

The simulation highlights a significant trade-off: the transition to a new steady state will involve a secular decline in the return on capital (r), which challenges the traditional mechanisms of wealth accumulation for savers. Furthermore, if the aging population leads to a reduction in national savings—as the Life Cycle Hypothesis suggests—or if the economy fails to achieve the target TFP growth of 1.1%, the buffer provided by capital intensification will dissipate.

Ultimately, the model suggests that demography is not necessarily destiny, but it is a formidable constraint. The sustainability of the Spanish economic model does not depend on reversing population trends, which is unlikely in the short term, but on pulling the two levers that remain within policy control: incentivizing productive investment to counteract the fall in savings, and fostering an ecosystem where technological efficiency compensates for the scarcity of human capital.

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